ZEROS OF OPTIMAL POLYNOMIAL APPROXIMANTS IN H^p

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ABSTRACT. For every $1 < p < \infty$ with $p \neq 2$, we construct a polynomial g which does not vanish in the closed unit disc and whose optimal linear polynomial approximant to $1/g$ in the Hardy space H^p vanishes in the open unit disc. This answers a question of Centner in the negative for every $p \neq 2$. Our approach reduces the location of the zero to the sign of a single boundary integral, which is positive for every target when $p = 2$. We also construct a single outer target whose optimal polynomial approximants vanish in the open unit disc for infinitely many degrees, and we show that every zero a of an optimal polynomial approximant in H^p to $1/f$ with $f(0) \neq 0$ satisfies $|a| > 2\sqrt{p-1}/p$, which for $p = 2$ recovers the classical fact that such zeros lie outside the closed unit disc. Conversely, targets concentrated at two points of the unit circle produce zeros at depth of order $(p-2)^2$, so the order of this bound is sharp as $p \to 2$. For $p = 4$ we determine the extremal radius exactly: every such zero satisfies $|a| > \sqrt{7/8}$, and this is sharp.

1. INTRODUCTION

Let H^p denote the classical Hardy space of analytic functions in the unit disc $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$, which for $0 < p < \infty$ is equipped with

$$\|f\|_{H^p}^p = \lim_{r \to 1^-} \int_0^{2\pi} |f(re^{i\theta})|^p \, \frac{\mathrm{d}\theta}{2\pi}.$$

Suppose that $1 < p < \infty$, that f is a nonzero function in H^p, and that $n \geq 0$ is an integer. Since L^p is uniformly convex and $q \mapsto qf$ is injective on the polynomials of degree at most n, there is a unique such polynomial q_n which minimizes $\|1 - qf\|_{H^p}$. Following the literature, we call q_n the *optimal polynomial approximant* (OPA) to $1/f$ of degree n. We do not assume that $1/f$ is in H^p; the definition only concerns the minimization problem.

As in the work of Cheng, Ross and Seco [7], of Cheng and Felder [6], and of Centner, Cheng and Felder [5], we write $q_{n,p}[f]$ for this polynomial, and we describe it by Birkhoff–James orthogonality: $q_{n,p}[f]$ is the OPA if and only if $q_{n,p}[f]f - 1$ is orthogonal in H^p to every multiple qf with $\deg q \leq n$ (Lemma 1). For $p = 2$ this is ordinary orthogonality. Our results show that

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the nonlinearity of the duality map $E \mapsto |E|^{p-2}E$ for $p \neq 2$ is exactly what allows zeros in $\mathbb{D}$.

For $p = 2$, the OPAs to $1/f$ do not vanish in the closed unit disc whenever $f(0) \neq 0$. This was first established by Chui [9], and later by Bénéteau, Khavinson, Liaw, Seco and Sola [2] by means of orthogonal polynomials. For $p \neq 2$, OPAs in H^p have been studied by Centner [3], by Centner, Cheng and Felder [5], and by Cheng and Felder [6]. Centner [3, Conjecture 7.1] conjectured that the OPAs to $1/f$ in H^p do not vanish in $\mathbb{D}$ whenever $1 < p < \infty$ and $f(0) \neq 0$, motivated by the design of stable digital filters [4, Section 4.4], and asked [3, Open Question 3] for an equivalent condition at degree one when $p \geq 2$. He also showed [3, Theorem 5.3] that the zeros of the OPAs to $1/f$ lie outside a disc whose radius depends on f. Bénéteau, Cheng, Felder, Khavinson, Manolaki and Maronikolakis [1, Section 4] recently proposed an approach to this question based on comparing $\|1 - Jf\|_{H^p}$ with $\|1 - cf\|_{H^p}$ for inner functions J and unimodular constants c. In the sequence spaces ℓ^p_A, Cheng, Ross and Seco [7] showed that OPAs may vanish in $\mathbb{D}$ for every $p \neq 2$.

The main purpose of the present paper is to show that the same is true in H^p, even for outer polynomial targets and for OPAs of degree one. Let $\mathcal{P}_1$ denote the space of polynomials of degree at most one.

Theorem 1. *Suppose that $1 < p < \infty$ and $p \neq 2$. There is a polynomial g which does not vanish in $\overline{\mathbb{D}}$ such that the OPA to $1/g$ of degree one in H^p has a zero in $\mathbb{D}$.*

In particular, Theorem 1 answers Centner's question in the negative for every $p \neq 2$. For $p > 2$, it also answers [3, Open Question 3] in the negative. For $p = 4$, an explicit polynomial g of degree 16 with integer coefficients is given in Section 9.2. Centner, Cheng and Felder [5, Theorems 5.13 and 5.16] showed that the zero of the OPA of degree one is bounded away from the origin by a constant $\rho > 0$ depending only on p, when the target is inner or when $p > 2$ is an even integer, and they asked whether one can take $\rho = 1$. Theorem 1 shows that one cannot take $\rho = 1$ uniformly over all targets when $p > 2$ is an even integer, and more generally for any $p \neq 2$; since our targets are polynomials, it says nothing about inner targets. In Section 9.4 we explain how it also yields counterexamples to [1, Conjecture 4.3].

Since composition with z^n commutes with the problem, we also obtain zeros for OPAs of every degree.

Corollary 1. *Suppose that $1 < p < \infty$, that $p \neq 2$, and that $n \geq 1$ is an integer. There is a polynomial g which does not vanish in $\overline{\mathbb{D}}$ such that the OPA to $1/g$ of degree n in H^p has n zeros in $\mathbb{D}$.*

In Corollary 1 the target depends on n. Our second result shows that a single target suffices for infinitely many degrees.

Theorem 2. *Suppose that $1 < p < \infty$ and $p \neq 2$. There is an outer function f in H^p with $f(0) \neq 0$ and integers $1 \leq a_1 < b_1 < a_2 < b_2 < \cdots$ with*

$b_k/a_k \to \infty$ such that the OPA to $1/f$ of degree n in H^p has a zero in $\mathbb{D}$ for every n in $\bigcup_k [a_k, b_k]$.

In particular, the set of such n has upper density one. The target f cannot be analytic and zero-free in a neighborhood of $\overline{\mathbb{D}}$; see Section 9.5.

Our third result shows that the zeros cannot lie far inside $\mathbb{D}$, and that they are pushed towards $\mathbb{T}$ as $p \to 2$.

Theorem 3. *Suppose that $1 < p < \infty$, that f is a function in H^p with $f(0) \neq 0$, and that $n \geq 1$ is an integer. If the OPA to $1/f$ of degree n vanishes at a, then*

$$|a| > \frac{2\sqrt{p-1}}{p}.$$

For $p = 2$, Theorem 3 is the result of Chui and of Bénéteau, Khavinson, Liaw, Seco and Sola mentioned above, and the proof given in Section 6 is short also in this case. In contrast to [3, Theorem 5.3], the radius in Theorem 3 does not depend on f. For $p \neq 2$, it gives an explicit value of the constant in [5, Theorems 5.13 and 5.16], for every target and every $1 < p < \infty$. This answers the request at the end of [5, Section 5] to extend these results to all $f \in H^p$ with $f(0) \neq 0$ and all $1 < p < \infty$. The hypothesis $f(0) \neq 0$ cannot be dropped: if $f(0) = 0$, then $\|1 - qf\|_{H^p} \geq |1 - q(0)f(0)| = 1$ for every polynomial q, so the OPA is identically zero. Note that the bound is invariant under $p \mapsto p/(p-1)$, and that $2\sqrt{p-1}/p = 1 - (p-2)^2/8 + O(|p-2|^3)$.

For $p \neq 2$, let $\rho_1(p)$ denote the infimum of $|a|$ over all zeros a of OPAs to $1/f$ of degree one in H^p, where f ranges over H^p with $f(0) \neq 0$. Theorems 1 and 3 show that $2\sqrt{p-1}/p \leq \rho_1(p) < 1$. Our fourth result shows that the order of the lower bound is correct as $p \to 2$.

Theorem 4. *Let $\kappa = (\pi + 2\log 2)/16$. Then*

$$\frac{\kappa^2}{2} \leq \liminf_{p \to 2} \frac{1 - \rho_1(p)}{(p-2)^2} \leq \limsup_{p \to 2} \frac{1 - \rho_1(p)}{(p-2)^2} \leq \frac{1}{8}.$$

The upper bound is Theorem 3. For the lower bound we use targets which are concentrated near two conjugate points of $\mathbb{T}$; they are not outer. For $p = 4$ the same targets turn out to be extremal.

Theorem 5. *Suppose that $f \in H^4$ with $f(0) \neq 0$ and that $n \geq 1$. If the OPA to $1/f$ of degree n in H^4 vanishes at a, then $|a| > \sqrt{7/8}$. Moreover, $\rho_1(4) = \sqrt{7/8}$.*

Theorem 3 gives only $|a| > \sqrt{3}/2$ for $p = 4$.

Degree one is the decisive case for this question. Indeed, suppose that the OPA q to $1/f$ of degree n has a zero $a \in \mathbb{D}$, and write $q = (z - a)P$. The minimal error is strictly smaller than 1, since $\|1 - cf\|_{H^p}^p = 1 - p\,\mathrm{Re}\big(cf(0)\big) + o(|c|)$ as $c \to 0$. As $\|1 - qf\|_{H^p} \geq |1 - q(0)f(0)|$, this forces $q(0) \neq 0$, so the function $F = Pf$ satisfies $F(0) \neq 0$. Since rP has degree at most n for every $r \in \mathcal{P}_1$, the polynomial $z - a$ is the OPA to $1/F$ of degree one. This

reduction can also be found in [3, Proposition 5.4], [6, Proposition 3.2.2] and [5, Proposition 4.1].

Let us briefly describe our approach. Let ϕ be a polynomial with $\phi(1) = 1$, and consider the targets

$$f_\alpha(z) = \frac{\phi(J_\alpha(z))}{1+\alpha z}, \qquad J_\alpha(z) = \frac{z+\alpha}{1+\alpha z},$$

as $\alpha \to 1^-$. Under the change of variables $w = J_\alpha(z)$, the problem of finding the OPA of degree one concentrates at the point $w = 1$, and in the limit it becomes the problem of minimizing the convex function

$$\Psi_\phi(v) = \int_{\mathbb{T}} |1-w|^{p-2} |R_\phi(w) + v\phi(w)|^p \, \mathrm{d}m(w), \qquad R_\phi(w) = \frac{1-\phi(w)}{1-w},$$

over $v \in \mathbb{C}$. The OPA has a zero in $\mathbb{D}$ precisely when the minimizer satisfies $\operatorname{Re} v > 1/2$. A short computation shows that the derivative of Ψ_ϕ in the direction of the real axis at the line $\operatorname{Re} v = 1/2$ is an integral of the real part of the analytic function $1/2 + wR_\phi(w)$, which takes the value $1/2$ at the origin, against the weight $|1-w|^{p-2}$ times a power of $|R_\phi + v\phi|$. For $p = 2$ the weight is constant, and the derivative is positive for every ϕ. For $p \neq 2$ the weight is not constant, and we exhibit a family of zero-free profiles ϕ for which the derivative is negative.

Organization. The present paper is organized into eight additional sections. In Section 2 we describe the OPA by Birkhoff–James orthogonality, introduce the limit problem, and show that it determines the location of the zero of the OPA to $1/f_\alpha$ for α close to 1. Section 3 contains the boundary identity, which reduces the location of the zero to the sign of a single integral. In Section 4 we choose the profiles and prove Theorem 1 and Corollary 1. In Section 5 we localize the argument of Section 2 at high degree and prove Theorem 2. Section 6 contains the proof of Theorem 3. In Section 7 we concentrate the target at two points of $\mathbb{T}$ and prove Theorem 4. Section 8 contains the proof of Theorem 5, and Section 9 contains some concluding remarks.

2. Orthogonality and the limit problem

Throughout the paper we write m for the normalized Lebesgue arc length measure on the unit circle $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$, we set $\gamma = p - 2$, and we write $\mathcal{P}_n$ for the polynomials of degree at most n. Recall that $\|f\|_{H^p}$ is the $L^p(\mathbb{T})$ norm of the boundary values of f.

2.1. Orthogonality and the OPA.

Let X be a normed space. Following Birkhoff and James, we say that $x \in X$ is orthogonal to $y \in X$, and write $x \perp_X y$, if $\|x + cy\| \geq \|x\|$ for every $c \in \mathbb{C}$. For $\alpha = |\alpha| e^{i\vartheta}$ and $s > 0$ we use the notation $\alpha^{\langle s \rangle} = |\alpha|^s e^{-i\vartheta}$ of [5], with $0^{\langle s \rangle} = 0$. The following characterization is due to James [8]; see also [5, Theorem 2.1].

Lemma 1. *Suppose that $1 < p < \infty$, and let μ be a positive measure. Then $x \perp y$ in $L^p(\mu)$ if and only if $\int x^{\langle p-1\rangle} y \, \mathrm{d}\mu = 0$. Consequently, if $f \in H^p$ is nonzero and $n \geq 0$, then a polynomial $q \in \mathcal{P}_n$ is the OPA $q_{n,p}[f]$ if and only if, with $E = 1 - qf$,*

$$\int_{\mathbb{T}} E^{\langle p-1\rangle} \, q' f \, \mathrm{d}m = 0 \qquad \text{for every } q' \in \mathcal{P}_n. \tag{1}$$

Proof. The function $c \mapsto \|x + cy\|^p_{L^p(\mu)}$ is convex and continuously differentiable on $\mathbb{C}$, with derivative at $c = 0$ in the direction of $e^{i\vartheta}$ equal to $p \operatorname{Re}(e^{i\vartheta} \int x^{\langle p-1\rangle} y \, \mathrm{d}\mu)$. So $c = 0$ is a minimizer if and only if this integral vanishes. The second assertion follows since q minimizes $\|1 - qf\|_{H^p}$ over $\mathcal{P}_n$ if and only if $E \perp q'f$ for every $q' \in \mathcal{P}_n$. □

Since $E^{\langle p-1\rangle} = |E|^{p-2}\overline{E}$, taking complex conjugates in (1) gives

$$\int_{\mathbb{T}} |E|^{p-2} E \, \overline{q'f} \, \mathrm{d}m = 0.$$

We will also use two standard estimates for H^p functions and polynomials.

Lemma 2. *Suppose that $1 < p < \infty$.*

1. *If $h \in H^p$ and $a \in \mathbb{D}$, then $|h(a)|^p \leq (1 + |a|)(1 - |a|)^{-1} \|h\|^p_{H^p}$.*
2. *(Nikolskii) There is $K = K_p$ such that $\|D\|_\infty \leq K N^{1/p} \|D\|_{H^p}$ for every $D \in \mathcal{P}_N$, where $\|D\|_\infty$ is the maximum of $|D|$ on $\mathbb{T}$.*

Proof. For (a), note that $|h|^p$ is subharmonic, so $|h(a)|^p$ is at most the Poisson integral of $|h|^p$ at a, and the Poisson kernel is bounded by $(1 + |a|)/(1 - |a|)$.

For (b), write $D = D * \mathcal{D}_N$, where $\mathcal{D}_N(e^{it}) = \sum_{k=0}^N e^{ikt}$ satisfies $|\mathcal{D}_N(e^{it})| \leq \min(N + 1, \pi/|t|)$, and apply Hölder's inequality; the $L^{p'}$ norm of $\mathcal{D}_N$ is at most $KN^{1/p}$. □

2.2. **The limit problem.** Let ϕ be a polynomial with $\phi(1) = 1$ and $\phi(0) \neq 0$. Since $\phi(1) = 1$, the function R_ϕ from the introduction is a polynomial, and we set $L_v = R_\phi + v\phi$ for $v \in \mathbb{C}$. Since $\gamma > -1$, the weight $|1 - w|^\gamma$ is integrable on $\mathbb{T}$, so $\Psi_\phi(v)$ is finite for every $v \in \mathbb{C}$.

Lemma 3. *The function Ψ_ϕ is strictly convex and coercive on $\mathbb{C}$, so it has a unique minimizer v_*. In the weighted space $L^p_\gamma = L^p(|1 - w|^\gamma \, \mathrm{d}m)$, the minimizer is characterized by the orthogonality relation $L_{v_*} \perp \phi$. Moreover, Ψ_ϕ is continuously differentiable with respect to the real coordinates $v = x + iy$, and*

$$\partial_x \Psi_\phi(v) = p \int_{\mathbb{T}} |1 - w|^\gamma \operatorname{Re}(L_v^{\langle p-1\rangle} \phi) \, \mathrm{d}m. \tag{2}$$

If ϕ has real coefficients, then v_* is real.

Proof. The map $v \mapsto L_v$ is affine with direction $\phi \not\equiv 0$, the function $\zeta \mapsto |\zeta|^p$ is strictly convex on $\mathbb{C}$, and the weight is positive almost everywhere. This gives strict convexity. Write $\|\cdot\|$ for the norm of $L^p(|1-w|^\gamma \, \mathrm{d}m)$. Minkowski's inequality gives $\Psi_\phi(v)^{1/p} \geq |v|\|\phi\| - \|R_\phi\|$, which is coercivity. Since $\Psi_\phi(v) =$

$\|R_\phi + v\phi\|_{L^p_\gamma}^p$, the point v_* minimizes Ψ_ϕ if and only if $L_{v_*} \perp \phi$ in L^p_γ, by the definition of orthogonality. For $p > 1$ the function $v \mapsto |a + vb|^p$ is continuously differentiable, with x-derivative $p\,\mathrm{Re}\big((a + vb)^{\langle p-1\rangle} b\big)$ bounded by $p|a + vb|^{p-1}|b|$. With $a = R_\phi(w)$ and $b = \phi(w)$, this is bounded uniformly in $w \in \mathbb{T}$ for v in compact sets, so we may differentiate under the integral sign. If ϕ has real coefficients, then so has R_ϕ, and since $w \mapsto \overline{w}$ preserves m we get $\Psi_\phi(\overline{v}) = \Psi_\phi(v)$. Uniqueness then gives $v_* = \overline{v_*}$. □

For $0 < \alpha < 1$, we set $C_\alpha = 1 - \alpha^2$ and define J_α and f_α as in the introduction. Each f_α is a rational function with poles outside $\overline{\mathbb{D}}$, and $f_\alpha(0) = \phi(\alpha) \to 1$ as $\alpha \to 1^-$. The next result shows that the limit problem decides on which side of $\mathbb{T}$ the zero of the OPA lies.

Lemma 4. *Suppose that* $\mathrm{Re}\, v_* > 1/2$. *Then the OPA to* $1/f_\alpha$ *of degree one has a zero in* $\mathbb{D}$ *for every* $\alpha < 1$ *sufficiently close to* 1. *If instead* $\mathrm{Re}\, v_* < 1/2$, *then it has no zeros in* $\overline{\mathbb{D}}$ *for every* $\alpha < 1$ *sufficiently close to* 1.

Proof. Step 1: a change of variables. Every $Q \in \mathcal{P}_1$ can be written as $Q = B(z+\alpha) + C_\alpha(1-v)$ with $B, v \in \mathbb{C}$. For $|z| = 1$ and $w = J_\alpha(z)$, we have $(z+\alpha)/(1+\alpha z) = w$ and $C_\alpha/(1+\alpha z) = 1-\alpha w$. Hence $1 - Qf_\alpha = E_\alpha(B, v, w)$, where

$$E_\alpha(B, v, w) = 1 - \big(Bw + (1-v)(1-\alpha w)\big)\phi(w).$$

Since J_α maps m to the harmonic measure for α, we have $\int_{\mathbb{T}} H \circ J_\alpha \,\mathrm{d}m = C_\alpha \int_{\mathbb{T}} H(w)|w-\alpha|^{-2}\,\mathrm{d}m(w)$. With $\mu_\alpha = |w-\alpha|^{-2}\,\mathrm{d}m(w)$, this yields the exact identity

$$\frac{\|1 - Qf_\alpha\|_{H^p}^p}{C_\alpha} = \int_{\mathbb{T}} |E_\alpha(B, v, w)|^p \,\mathrm{d}\mu_\alpha(w). \tag{3}$$

We will use that $\mu_\alpha(\mathbb{T}) = 1/C_\alpha$, that $|w-\alpha|^2 = (1-\alpha)^2 + \alpha|1-w|^2 \geq \alpha|1-w|^2$, and that

$$E_\alpha(1, v, w) = (1-w)L_v(w) - (1-\alpha)(1-v)w\phi(w). \tag{4}$$

The last identity follows from the fact that $1 - \big(1 - v(1-w)\big)\phi = (1-\phi) + v(1-w)\phi = (1-w)L_v$.

Step 2: competitors. We first consider the polynomials $Q_{\alpha,v} = z + \alpha + C_\alpha(1-v)$. In $L^p(\mu_\alpha)$, the last term of (4) has norm at most $|1-v|\|\phi\|_\infty(1-\alpha)C_\alpha^{-1/p}$, which tends to 0. For $\alpha \geq 1/2$, the pth power of the first term is bounded by $2|1-w|^\gamma|L_v|^p$ with respect to μ_α, and it converges pointwise to $|1-w|^\gamma|L_v|^p$ for $w \neq 1$. Dominated convergence and Minkowski's inequality give

$$\lim_{\alpha\to 1^-} \frac{\|1 - Q_{\alpha,v} f_\alpha\|_{H^p}^p}{C_\alpha} = \Psi_\phi(v) \qquad \text{and} \qquad \|1 - Q_{\alpha,1} f_\alpha\|_{H^p}^p \leq 2C_\alpha \Psi_\phi(1), \tag{5}$$

where the inequality holds for $\alpha \geq 1/2$ since the last term of (4) vanishes when $v = 1$.

Step 3: the OPA stays bounded. Let $Q_\alpha = A_\alpha + B_\alpha z$ be the OPA $q_{1,p}[f_\alpha]$, set $r_\alpha = 1 - Q_\alpha f_\alpha$, and define v_α by $A_\alpha = \alpha B_\alpha + C_\alpha(1 - v_\alpha)$. Since Q_α is optimal, (5) gives $\|r_\alpha\|_{H^p}^p \le 2C_\alpha \Psi_\phi(1)$, and so $r_\alpha \to 0$ locally uniformly in $\mathbb{D}$ by Lemma 2 (a). Since $J_\alpha - 1 = -(1-\alpha)(1-z)/(1+\alpha z)$, we also have $f_\alpha \to 1/(1+z)$ locally uniformly, and so $Q_\alpha = (1 - r_\alpha)/f_\alpha \to 1 + z$. Evaluating at 0 and 1/2, we find that $A_\alpha \to 1$ and $B_\alpha \to 1$. At $z = -\alpha$ we have $f_\alpha(-\alpha) = \phi(0)/C_\alpha$ and $Q_\alpha(-\alpha) = C_\alpha(1 - v_\alpha)$, so

$$\bigl|1 - \phi(0)(1 - v_\alpha)\bigr|^p = |r_\alpha(-\alpha)|^p \le \frac{1+\alpha}{1-\alpha}\, 2C_\alpha \Psi_\phi(1) \le 8\Psi_\phi(1).$$

Since $\phi(0) \neq 0$, the numbers v_α are bounded.

Step 4: identification of the limit. Suppose that $\alpha_j \to 1^-$ and that $v_{\alpha_j} \to v$. For $w \neq 1$, the integrand in (3) with $(B, v) = (B_{\alpha_j}, v_{\alpha_j})$ converges to $|1 - w|^\gamma |L_v(w)|^p$. By Fatou's lemma, the optimality of Q_{α_j}, and (5) with $v = v_*$,

$$\Psi_\phi(v) \le \liminf_{j\to\infty} \frac{\|r_{\alpha_j}\|_{H^p}^p}{C_{\alpha_j}} \le \limsup_{j\to\infty} \frac{\|1 - Q_{\alpha_j, v_*} f_{\alpha_j}\|_{H^p}^p}{C_{\alpha_j}} = \Psi_\phi(v_*).$$

Hence $v = v_*$ by uniqueness, and since the limit point was arbitrary, $v_\alpha \to v_*$.

Step 5: location of the zero. Inserting $A_\alpha = \alpha B_\alpha + C_\alpha(1 - v_\alpha)$, we get

$$\frac{|A_\alpha|^2 - |B_\alpha|^2}{C_\alpha} = -|B_\alpha|^2 + 2\alpha \operatorname{Re}\bigl(B_\alpha \overline{(1 - v_\alpha)}\bigr) + C_\alpha |1 - v_\alpha|^2 \longrightarrow 1 - 2\operatorname{Re} v_*.$$

If $\operatorname{Re} v_* > 1/2$, then $|A_\alpha| < |B_\alpha|$ for α close to 1, and the zero $-A_\alpha/B_\alpha$ of Q_α lies in $\mathbb{D}$. If $\operatorname{Re} v_* < 1/2$, then $|A_\alpha| > |B_\alpha|$, and Q_α has no zeros in $\overline{\mathbb{D}}$. □

3. A boundary identity

In view of Lemma 4, we need to decide on which side of the line $\operatorname{Re} v = 1/2$ the minimizer of Ψ_ϕ lies. The following identity shows that the derivative across this line has a particularly simple form.

Lemma 5. *Let $h_\phi(w) = 1/2 + wR_\phi(w)$, which is a polynomial with $h_\phi(0) = 1/2$. For $|w| = 1$ and $y \in \mathbb{R}$,*

$$\operatorname{Re}\bigl(\overline{L_{1/2+iy}(w)}\, \phi(w)\bigr) = \operatorname{Re} h_\phi(w).$$

Equivalently, $\operatorname{Re}\bigl(L_{1/2+iy}^{\langle p-1\rangle}\phi\bigr) = |L_{1/2+iy}|^{p-2} \operatorname{Re} h_\phi$ *on* $\mathbb{T}$. *Consequently,*

$$\partial_x \Psi_\phi(1/2 + iy) = p \int_{\mathbb{T}} |1 - w|^\gamma |L_{1/2+iy}|^{p-2} \operatorname{Re} h_\phi \, \mathrm{d}m. \tag{6}$$

Proof. The term $iy|\phi|^2$ is purely imaginary, so we may assume that $y = 0$. Write $R = R_\phi$. Inserting $\phi = 1 - (1 - w)R$, we get $\overline{R}\phi = \overline{R} - (1 - w)|R|^2$ and $|\phi|^2 = 1 - 2\operatorname{Re}\bigl((1 - w)R\bigr) + |1 - w|^2|R|^2$. Since $|1 - w|^2 = 2\operatorname{Re}(1 - w)$ on $\mathbb{T}$, it follows that

$$\operatorname{Re}\bigl(\overline{R}\phi + |\phi|^2/2\bigr) = \operatorname{Re} R + \frac{1}{2} - \operatorname{Re}\bigl((1 - w)R\bigr) = \frac{1}{2} + \operatorname{Re}(wR).$$

Since $L^{\langle p-1\rangle} = |L|^{p-2}\overline{L}$, the formula (6) is (2) with $v = 1/2 + iy$. □

Lemma 6. *If $\partial_x \Psi_\phi(1/2 + iy) < 0$ for every $y \in \mathbb{R}$, then $\operatorname{Re} v_* > 1/2$. If $\partial_x \Psi_\phi(1/2 + iy) > 0$ for every $y \in \mathbb{R}$, then $\operatorname{Re} v_* < 1/2$. If ϕ has real coefficients, then $v_* > 1/2$ if and only if $\Psi'_\phi(1/2) < 0$.*

Proof. Write $v_* = x_* + iy_*$. The function $x \mapsto \Psi_\phi(x + iy_*)$ is convex and continuously differentiable, and its derivative vanishes at x_*. Since the derivative is nondecreasing, a negative value at $x = 1/2$ forces $x_* > 1/2$, and a positive value forces $x_* < 1/2$. If ϕ has real coefficients, then v_* is real by Lemma 3, and strict convexity on $\mathbb{R}$ gives the converse. □

For $p = 2$ the weight in (6) is constant, and we recover the fact that there are no interior zeros in the limit problem.

Corollary 2. *Suppose that $p = 2$. Then $\partial_x \Psi_\phi(1/2 + iy) = 2 \operatorname{Re} h_\phi(0) = 1$ for every ϕ and every $y \in \mathbb{R}$. In particular, $\operatorname{Re} v_* < 1/2$.*

For $p \neq 2$, the weight $|1 - w|^\gamma |L_{1/2}|^\gamma$ is not constant, and the sign of (6) is decided by how the weight correlates with $\operatorname{Re} h_\phi - 1/2$, which has mean 0. Our task is to choose ϕ so that this correlation wins.

4. The profiles and the proof of Theorem 1

Suppose that $p \neq 2$, let $\sigma = \operatorname{sign}(2 - p)$, and for $\lambda > 0$ set

$$\phi_\lambda(w) = 1 + \lambda(1 - w)(1 + \sigma w).$$

In other words, $\phi_\lambda(w) = 1 + \lambda(1 - w^2)$ if $p < 2$ and $\phi_\lambda(w) = 1 + \lambda(1 - w)^2$ if $p > 2$. In both cases $\phi_\lambda(1) = 1$ and $\phi_\lambda(0) = 1 + \lambda$. Our first observation is that these profiles produce targets without zeros.

Lemma 7. *The polynomial ϕ_λ does not vanish in $\overline{\mathbb{D}}$. Consequently, the function f_α built from ϕ_λ does not vanish in $\overline{\mathbb{D}}$.*

Proof. If $\sigma = 1$, then $\operatorname{Re}(1 - w^2) \geq 0$ for $w \in \overline{\mathbb{D}}$, so $\operatorname{Re} \phi_\lambda \geq 1$. If $\sigma = -1$, then $\phi_\lambda(w) = 0$ forces $1 - w = \pm i/\sqrt{\lambda}$, so $|w|^2 = 1 + 1/\lambda > 1$. The second assertion follows since J_α maps $\overline{\mathbb{D}}$ into itself and $1 + \alpha z \neq 0$ for $z \in \overline{\mathbb{D}}$. □

Set $G(w) = 1 + \sigma w$ and $P(w) = (1 + w)G(w)$. Then $R_{\phi_\lambda} = -\lambda G$, and with $\varepsilon = 1/\lambda$ we find that

$$L_{1/2} = \frac{1}{2} - \frac{\lambda}{2} P = -\frac{\lambda}{2}(P - \varepsilon) \qquad \text{and} \qquad \operatorname{Re} h_{\phi_\lambda} = \lambda\big(\varepsilon/2 - \operatorname{Re}(wG)\big).$$

Inserting this into (6), we obtain

$$\frac{\Psi'_{\phi_\lambda}(1/2)}{p\lambda(\lambda/2)^\gamma} = \int_{\mathbb{T}} |1 - w|^\gamma |P - \varepsilon|^\gamma \big(\varepsilon/2 - \operatorname{Re}(wG)\big)\, \mathrm{d}m. \tag{7}$$

As $\lambda \to \infty$, the right-hand side tends to $-F_\sigma$, where F_σ is the integral computed in the next lemma.

Lemma 8. *Let* $F_\sigma = \int_{\mathbb{T}} |1-w|^\gamma |P|^\gamma \operatorname{Re}(wG)\,\mathrm{d}m$, *and set*

$$I(a,b) = \frac{1}{\pi}\int_0^\pi \sin^a s\,|\cos s|^b\,\mathrm{d}s = \frac{1}{\pi}\mathrm{B}\big((a+1)/2,(b+1)/2\big).$$

Then

$$F_\sigma = -\sigma\gamma\,\frac{2^{3\gamma+1}}{3\gamma+4}\,I(\gamma,2\gamma+2).$$

In particular, $F_\sigma > 0$.

Proof. Suppose first that $\sigma = 1$, and write $w = e^{2is}$ with $|s| < \pi/2$. Then $\mathrm{d}m = \mathrm{d}s/\pi$, $|1-w| = 2|\sin s|$, $|P| = |1+w|^2 = 4\cos^2 s$, and

$$\operatorname{Re}(w+w^2) = 2\cos 3s\cos s = 2\big(4\cos^4 s - 3\cos^2 s\big).$$

Thus $F_1 = 2^{3\gamma+1}\big(4I(\gamma,2\gamma+4) - 3I(\gamma,2\gamma+2)\big)$. The identity $\mathrm{B}(x,y+1) = \mathrm{B}(x,y)\,y/(x+y)$ yields $I(\gamma,2\gamma+4) = I(\gamma,2\gamma+2)(2\gamma+3)/(3\gamma+4)$, and the formula follows. If $\sigma = -1$, then $P(w) = 1-w^2$ and $wG(w) = w-w^2$. Since $|1-w^2| = |1-w||1+w|$, the substitution $w \mapsto -w$ shows that $F_{-1} = -F_1$. Finally, $\sigma\gamma < 0$, $3\gamma+4 > 0$ and $I(\gamma,2\gamma+2) > 0$, since $\gamma > -1$. □

Armed with Lemma 8, we can now show that the minimizer of the limit problem lies on the correct side of the line $\operatorname{Re} v = 1/2$.

Lemma 9. *Suppose that* $p \neq 2$. *Then*

$$\lim_{\lambda\to\infty} \frac{\Psi'_{\phi_\lambda}(1/2)}{p\lambda(\lambda/2)^\gamma} = -F_\sigma < 0.$$

In particular, $v_* > 1/2$ *for* ϕ_λ *when* λ *is sufficiently large.*

Proof. The integrand in (7) converges to $-|1-w|^\gamma|P|^\gamma\operatorname{Re}(wG)$ for $w \neq -1$. We show that for $\lambda \geq 1$, so that $\varepsilon \leq 1$, it is bounded by a constant times $|1-w|^\gamma$, which is integrable. Note that $|P| \leq 4$ and $|wG| \leq 2$ on $\mathbb{T}$.

If $\gamma > 0$, then $|P-\varepsilon|^\gamma \leq 5^\gamma$ and $|\varepsilon/2 - \operatorname{Re}(wG)| \leq 5/2$.

If $-1 < \gamma < 0$, then $\sigma = 1$, and $P = (1+w)^2 = 4c^2 w$ with $c = \cos(\theta/2)$ and $w = e^{i\theta}$. We first show that $|P-\varepsilon| \geq (|P|+\varepsilon)/3$. If $\cos\theta \leq 0$, then $|P-\varepsilon|^2 = |P|^2 - 2\varepsilon|P|\cos\theta + \varepsilon^2 \geq (|P|+\varepsilon)^2/2$. If $\cos\theta > 0$, then $|P| = 2(1+\cos\theta) > 2$, and so $|P-\varepsilon| \geq |P|-1 \geq (|P|+1)/3$. Next, the inequality $|\cos 3s| \leq 3|\cos s|$ gives $|\operatorname{Re}(wG)| = 2|\cos(\theta/2)\cos(3\theta/2)| \leq 6c^2 = 3|P|/2$, so that $|\varepsilon/2 - \operatorname{Re}(wG)| \leq 3(|P|+\varepsilon)/2$. Since $\gamma < 0$, we conclude that

$$|P-\varepsilon|^\gamma|\varepsilon/2 - \operatorname{Re}(wG)| \leq \frac{3^{1-\gamma}}{2}(|P|+\varepsilon)^{1+\gamma} \leq \frac{3^{1-\gamma}}{2}\,5^{1+\gamma}.$$

In both cases, dominated convergence gives the limit, and its sign is given by Lemma 8. The final assertion follows from Lemma 6, since ϕ_λ has real coefficients. □

It remains to replace the rational targets f_α by polynomials. In preparation for this, we record that OPAs depend continuously on the target. For a polynomial q we write $\|q\|_\infty$ for the maximum of $|q|$ on $\mathbb{T}$.

Lemma 10. *Suppose that $n \geq 1$, that f_j and f are nonzero functions in H^p, and that $f_j \to f$ in H^p. Then the OPAs to $1/f_j$ of degree n converge coefficientwise to the OPA to $1/f$ of degree n. In particular, if the latter is not identically zero and has a zero in $\mathbb{D}$, then the OPAs to $1/f_j$ of degree n have a zero in $\mathbb{D}$ for every sufficiently large j.*

Proof. Let $\mathcal{P}_n$ denote the polynomials of degree at most n. Since $q \mapsto \|qf\|_{H^p}$ is a norm on $\mathcal{P}_n$ and all norms on $\mathcal{P}_n$ are equivalent, there is $c > 0$ such that $\|qf\|_{H^p} \geq c\|q\|_\infty$. As $\|q(f_j - f)\|_{H^p} \leq \|q\|_\infty \|f_j - f\|_{H^p}$, the same holds for f_j with $c/2$ when j is large. The OPA Q_j to $1/f_j$ has error at most $\|1\|_{H^p} = 1$, so $\|Q_j f_j\|_{H^p} \leq 2$, and the coefficients of Q_j are bounded. If $Q_{j_k} \to Q$, then

$$\|1 - Qf\|_{H^p} = \lim_{k\to\infty} \|1 - Q_{j_k} f_{j_k}\|_{H^p} \leq \lim_{k\to\infty} \|1 - Q' f_{j_k}\|_{H^p} = \|1 - Q' f\|_{H^p}$$

for every $Q' \in \mathcal{P}_n$, so Q is the OPA to $1/f$ by uniqueness. The final assertion follows from Hurwitz's theorem, which applies because the limiting polynomial is not identically zero. □

Proof of Theorem 1. Suppose that $p \neq 2$, and choose λ by Lemma 9 so that $v_* > 1/2$ for ϕ_λ. By Lemma 4, there is $\alpha \in (0, 1)$ such that the OPA to $1/f_\alpha$ of degree one has a zero in $\mathbb{D}$, and by Lemma 7 the function f_α does not vanish in $\overline{\mathbb{D}}$. Since f_α is analytic in the disc of radius $1/\alpha$, its Taylor polynomials g_N converge to f_α uniformly on $\overline{\mathbb{D}}$. By Lemma 10, the OPA to $1/g_N$ of degree one has a zero in $\mathbb{D}$ for large N. Moreover, g_N does not vanish in $\overline{\mathbb{D}}$ as soon as $|g_N - f_\alpha| < \min_{\overline{\mathbb{D}}} |f_\alpha|$ on $\overline{\mathbb{D}}$. □

Proof of Corollary 1. Let F be the polynomial from Theorem 1, let $z - a$ be a multiple of the OPA to $1/F$ of degree one, with $a \in \mathbb{D}$, and set $g(z) = F(z^n)$. Then g does not vanish in $\overline{\mathbb{D}}$. Let Q be the OPA to $1/g$ of degree n. If $\omega^n = 1$, then $g(\omega z) = g(z)$, and since rotations preserve m, the polynomial $Q(\omega z)$ is also optimal. By uniqueness $Q(\omega z) = Q(z)$, so $Q(z) = A + Bz^n$. The substitution $w = z^n$ preserves m, and so

$$\|1 - (A + Bz^n)g\|_{H^p} = \|1 - (A + Bw)F\|_{H^p}.$$

Hence $A + Bw$ is the OPA to $1/F$ of degree one, and $Q(z)$ is a multiple of $z^n - a$, which has n zeros of modulus $|a|^{1/n} < 1$. □

5. One target and infinitely many degrees

In this section we fix $p \neq 2$, choose λ by Lemma 9 so that $v_* > 1/2$ for $\phi = \phi_\lambda$, and fix a number θ with $1 - 1/p < \theta < 1$. Recall that ϕ has real coefficients, so v_* is real. For $0 < \alpha < 1$ we set $\delta = 1 - \alpha$, $\beta = 1 - \delta^\theta$ and

$$u_\alpha(z) = (1 + \beta z) f_\alpha(z) = \frac{(1 + \beta z)\, \phi(J_\alpha(z))}{1 + \alpha z}.$$

Then u_α is analytic in the disc of radius $1/\alpha$, it does not vanish in $\overline{\mathbb{D}}$ by Lemma 7, and $u_\alpha(0) = \phi(\alpha) > 0$. The factor $1 + \beta z$ cancels most of the pole

of f_α at $-1/\alpha$, and the next result shows that u_α is a small perturbation of 1.

Lemma 11. *As $\alpha \to 1^-$, we have $\|u_\alpha - 1\|_{H^p} \to 0$ and $\int_{\mathbb{T}} |\log|u_\alpha|| \, \mathrm{d}m \to 0$.*

Proof. We have

$$u_\alpha - 1 = \frac{1+\beta z}{1+\alpha z}(\phi(J_\alpha) - 1) + \frac{(\beta - \alpha)z}{1+\alpha z}.$$

Note that $0 < \alpha - \beta < \delta^\theta$, so $|1+\beta z| \le |1+\alpha z| + \delta^\theta$. Moreover, $|1 + \alpha e^{it}| \ge c(\delta + |\pi - |t||)$ for $|t| \le \pi$ and an absolute constant $c > 0$, and so $\int_{\mathbb{T}} |1+\alpha z|^{-p} \, \mathrm{d}m \le C_p \delta^{1-p}$. Since $J_\alpha(z) \to 1$ for $z \in \mathbb{T} \setminus \{-1\}$, dominated convergence gives $\|\phi(J_\alpha) - 1\|_{H^p} \to 0$. Hence

$$\|u_\alpha - 1\|_{H^p} \le \|\phi(J_\alpha) - 1\|_{H^p} + (\|\phi\|_\infty + 2) C_p^{1/p} \delta^{\theta - 1 + 1/p},$$

which tends to 0 since $\theta > 1 - 1/p$. For the second assertion, note that $\log|\phi \circ J_\alpha| \to 0$ boundedly almost everywhere, since ϕ does not vanish in $\overline{\mathbb{D}}$. Moreover, $|1 + rz|^2 = 1 + r^2 + 2r \operatorname{Re} z$ is affine in $\operatorname{Re} z$, and comparing the values at $\operatorname{Re} z = \pm 1$ we find that $|1+z|/2 \le |1+rz| \le 2$ for $z \in \mathbb{T}$ and $0 \le r \le 1$. By dominated convergence, $\log|1+rz| \to \log|1+z|$ in $L^1(\mathbb{T})$ as $r \to 1^-$, which we apply with $r = \alpha$ and $r = \beta$. □

The following result is the heart of the proof of Theorem 2. It is a version of Lemma 4 for OPAs of high degree, in which the polynomial part of the OPA first inverts the background $g(1+\beta z)$, and the zero is created at the scale δ around the point -1.

Lemma 12. *Suppose that g is analytic and does not vanish in a neighborhood of $\overline{\mathbb{D}}$, and set $F_\alpha = g u_\alpha$ and*

$$a(\alpha) = \delta^{-\theta} \log^2(1/\delta) \qquad \text{and} \qquad b(\alpha) = \frac{\delta^{-(1+\theta)/2}}{\log(1/\delta)}.$$

There is $\alpha_0 < 1$ such that for every $\alpha_0 < \alpha < 1$ and every integer N with $a(\alpha) \le N \le b(\alpha)$, the OPA to $1/F_\alpha$ of degree N has a zero in $\mathbb{D}$.

Proof. Suppose that the assertion is false. Then there are $\alpha_j \to 1^-$ and integers $N_j \in [a(\alpha_j), b(\alpha_j)]$ such that the OPA to $1/F_{\alpha_j}$ of degree N_j has no zeros in $\mathbb{D}$. We work along this sequence, suppress the index j, and write N, F and $C = C_\alpha = \delta(2-\delta)$. We will only use that

$$N\delta^\theta \ge \log^2(1/\delta) \qquad \text{and} \qquad N^2 \delta^{1+\theta} \le \frac{1}{\log^2(1/\delta)}. \tag{8}$$

Since $\theta < 1$, the second condition implies that $N\delta \to 0$. Let Q be the OPA to $1/F$ of degree N, set $r = 1 - QF$ and $S = g(1+\beta z)$, so that $F = S f_\alpha$, and let c_g denote the minimum of $|g|$ on $\overline{\mathbb{D}}$. Below, K denotes a constant which does not depend on α, and which may change from line to line.

Step 1: a lower bound for $|F|$. We first show that $|F| \ge K^{-1}$ on $\mathbb{T}$. For $z = e^{it}$, the ratio $|1+\beta z|^2/|1+\alpha z|^2$ is a quotient of two affine functions of $\cos t$, and is therefore monotone in $\cos t$. Its values at $\cos t = 1$ and

$\cos t = -1$ are $(1+\beta)^2/(1+\alpha)^2 \geq 1/4$ and $(1-\beta)^2/(1-\alpha)^2 \geq 1$. Hence $|1+\beta z| \geq |1+\alpha z|/2$, and so $|F| \geq c_g \min_{\overline{\mathbb{D}}}|\phi|/2$ on $\mathbb{T}$.

Step 2: a competitor. Let $m = \lfloor N/2 \rfloor$, let T_g be the Taylor polynomial of $1/g$ of degree $m-1$, and set $T = T_g \sum_{k=0}^{m-1}(-\beta z)^k$. Then $TS = T_g g(1-(-\beta z)^m)$. Since $1/g$ is analytic in a disc of radius larger than 1, there are K and $\rho < 1$ such that $|T_g g - 1| \leq K\rho^m$ on $\overline{\mathbb{D}}$. Moreover, $\beta^m \leq \exp(-\delta^\theta m) \leq 2\exp(-\log^2(1/\delta)/2)$ by (8). Hence $\varepsilon = \max_{\overline{\mathbb{D}}}|TS-1|$ tends to 0 faster than any power of δ. Let $\tilde{Q} = Q_{\alpha,v_*}T$, which has degree at most N. On $\mathbb{T}$ we have $|z+\alpha| = |1+\alpha z| \geq \delta$ and $C \leq 2\delta$, so $|Q_{\alpha,v_*}f_\alpha| \leq (1+2|1-v_*|)\|\phi\|_\infty$. Since

$$1 - \tilde{Q}F = (1 - Q_{\alpha,v_*}f_\alpha) - Q_{\alpha,v_*}f_\alpha(TS-1),$$

the optimality of Q, Minkowski's inequality and (5) yield

$$\limsup \frac{\|r\|_{H^p}^p}{C} \leq \Psi_\phi(v_*) \qquad \text{and} \qquad \|r\|_{H^p}^p + \|1-\tilde{Q}F\|_{H^p}^p \leq K\delta. \tag{9}$$

Step 3: the OPA is bounded on $\mathbb{T}$. We now show that $\|Q\|_\infty \leq K$. On $\mathbb{T}$ we have $|1+\beta z| = |z+\beta| \geq 1-\beta = \delta^\theta$, and $|\alpha + C(1-v_*) - \beta| \leq \delta^\theta + 2\delta|1-v_*|$. Hence $|Q_{\alpha,v_*}/S| \leq 3/c_g$ on $\mathbb{T}$ for α close to 1, and since $\tilde{Q} = (Q_{\alpha,v_*}/S)\,TS$, we get $\|\tilde{Q}\|_\infty \leq K$. The polynomial $D = Q - \tilde{Q}$ has degree at most N and satisfies $DF = (1-\tilde{Q}F) - r$, so $\|D\|_{H^p} \leq K\delta^{1/p}$ by (9) and Step 1. By Nikolskii's inequality (Lemma 2 (b)), $\|D\|_\infty \leq KN^{1/p}\delta^{1/p}$, and since $N\delta \to 0$, it follows that $\|D\|_\infty \to 0$. Hence $\|Q\|_\infty \leq K$.

Step 4: rescaling around -1. Set $\kappa = g(-1)\delta^{\theta-1}$ and $\hat{Q}(s) = \kappa Q(-1+\delta s)$. For $z = -1+\delta s$ we have $z+\alpha = \delta(s-1)$, $1+\alpha z = \delta(1+s-\delta s)$ and $1+\beta z = \delta^\theta(1+\beta\delta^{1-\theta}s)$, and therefore

$$\frac{F(-1+\delta s)}{\kappa} = \frac{g(-1+\delta s)}{g(-1)} \cdot \frac{1+\beta\delta^{1-\theta}s}{1+s-\delta s} \cdot \phi\Big(\frac{s-1}{1+s-\delta s}\Big). \tag{10}$$

As $\alpha \to 1^-$, this converges to $\phi(W(s))/(1+s)$ with $W(s) = (s-1)/(s+1)$, uniformly on compact subsets of $\mathbb{C} \setminus \{-1\}$.

Step 5: the rescaled OPA is asymptotically affine. Since Q has degree at most N, we have $|Q(z)| \leq |z|^N\|Q\|_\infty \leq e\|Q\|_\infty$ for $1 \leq |z| \leq 1+1/N$. Write $\hat{Q}(s) = \sum_k c_k(s-2)^k$. The Cauchy estimates for Q on the disc of radius $1/(2N)$ centered at $-1+2\delta$ give $|c_k| \leq K\delta^{\theta-1}(2N\delta)^k$. For $|s-2| \leq R$ and α so close to 1 that $2N\delta R \leq 1/2$, we get

$$|\hat{Q}(s) - c_0 - c_1(s-2)| \leq K\delta^{\theta-1}(N\delta R)^2 = KR^2N^2\delta^{1+\theta},$$

which tends to 0 by (8). Next, we show that c_0 and c_1 are bounded. If $|s-2| \leq 1/2$ and $z = -1+\delta s$, then $1-|z|^2 = 2\delta\operatorname{Re} s - \delta^2|s|^2 \geq 2\delta$ for small δ, and Lemma 2 (a) and (9) show that r is bounded there. The three factors in (10) are bounded away from 0 for such s, since $J_\alpha(z) \in \mathbb{D}$ and ϕ does not vanish in $\overline{\mathbb{D}}$. Hence $\hat{Q} = \kappa(1-r)/F$ is bounded on $|s-2| \leq 1/2$, and the Cauchy estimates bound c_0 and c_1.

Step 6: identification of the limit. Passing to a subsequence, we may assume that $\hat{Q}(s) \to A + Bs$ uniformly on compact subsets of $\mathbb{C}$. Fix $w \in \mathbb{T} \setminus \{1\}$, and let $z \in \mathbb{T}$ be the point with $J_\alpha(z) = w$. Then

$$z = -1 + \delta s_\alpha, \qquad s_\alpha = \frac{1+w}{1-\alpha w} \to s_w = \frac{1+w}{1-w}.$$

Since $W(s_w) = w$ and $1 + s_w = 2/(1-w)$, the convergence of $\hat{Q}$ and of (10) gives

$$r(z) = 1 - \hat{Q}(s_\alpha)\frac{F(z)}{\kappa} \to 1 - \ell(w)\phi(w), \qquad \ell(w) = \frac{A(1-w) + B(1+w)}{2}.$$

By the change of variables in the proof of Lemma 4, we have $\|r\|_{H^p}^p / C = \int_{\mathbb{T}} |r(J_\alpha^{-1}(w))|^p \, \mathrm{d}\mu_\alpha(w)$. Fatou's lemma and (9) give

$$\int_{\mathbb{T}} \frac{|1 - \ell(w)\phi(w)|^p}{|1-w|^2} \, \mathrm{d}m(w) \le \liminf \frac{\|r\|_{H^p}^p}{C} \le \Psi_\phi(v_*).$$

Since $\ell(1)\phi(1) = B$ and $|1-w|^{-2}$ is not integrable on $\mathbb{T}$, the left-hand side is finite only if $B = 1$. Then $\ell(w) = 1 - v(1-w)$ with $v = (1-A)/2$, and the identity $1 - (1 - v(1-w))\phi = (1-w)L_v$ shows that the left-hand side is $\Psi_\phi(v)$. Hence $v = v_*$ by uniqueness. Since the limit does not depend on the subsequence, we conclude that $\hat{Q}(s) \to s + 1 - 2v_*$ uniformly on compact subsets of $\mathbb{C}$.

Step 7: the zero. The limit has the zero $s_0 = 2v_* - 1 > 0$. By Hurwitz's theorem, $\hat{Q}$ has a zero s_δ with $s_\delta \to s_0$, and so Q vanishes at $z = -1 + \delta s_\delta$, where $|z|^2 = 1 - 2\delta \operatorname{Re} s_\delta + \delta^2 |s_\delta|^2 < 1$ for small δ. This contradicts the choice of the sequence. □

Note that $b(\alpha)/a(\alpha) = \delta^{-(1-\theta)/2} \log^{-3}(1/\delta) \to \infty$ as $\alpha \to 1^-$. We are now ready to prove Theorem 2 by a diagonal argument.

Proof of Theorem 2. For α close to 1 let $I(\alpha)$ denote the set of integers in $[a(\alpha), b(\alpha)]$. Set $g_0 = 1$. We will choose $\alpha_k \to 1^-$ and $\eta_k > 0$ recursively, and set $g_k = g_{k-1} u_{\alpha_k}$. Each g_k is analytic and does not vanish in a neighborhood of $\overline{\mathbb{D}}$.

Suppose that g_{k-1} and $\eta_1, \dots, \eta_{k-1}$ have been chosen, and set $\varepsilon_k = 2^{-k-1} \min(1, \eta_1, \dots, \eta_{k-1})$. By Lemmas 11 and 12, we may choose α_k such that Lemma 12 applies to $g = g_{k-1}$ and $\alpha = \alpha_k$, such that $\min I(\alpha_k) > \max I(\alpha_{k-1})$, and such that

$$\max_{\overline{\mathbb{D}}} |g_{k-1}| \cdot \|u_{\alpha_k} - 1\|_{H^p} \le \varepsilon_k \qquad \text{and} \qquad \int_{\mathbb{T}} |\log |u_{\alpha_k}|| \, \mathrm{d}m \le 2^{-k}.$$

Then the OPA to $1/g_k$ of degree N has a zero in $\mathbb{D}$ for every $N \in I(\alpha_k)$, and it is not identically zero, since $g_k(0) > 0$. Since $I(\alpha_k)$ is finite, Lemma 10 provides $\eta_k > 0$ such that the same holds for the OPAs to $1/h$ whenever $h \in H^p$ and $\|h - g_k\|_{H^p} < \eta_k$.

Since $\|g_k - g_{k-1}\|_{H^p} \le \varepsilon_k$, the sequence (g_k) converges in H^p to a function f, and

$$\|f - g_k\|_{H^p} \le \sum_{j>k} \varepsilon_j \le \eta_k \sum_{j>k} 2^{-j-1} < \eta_k.$$

Hence the OPA to $1/f$ of degree N has a zero in $\mathbb{D}$ for every N in the infinite set $\bigcup_k I(\alpha_k)$. Moreover, $|f(0)-1| \le \|f-1\|_{H^p} \le \sum_j \varepsilon_j \le 1/2$. Finally, each u_{α_k} is outer with $u_{\alpha_k}(0) > 0$, so

$$g_k(z) = \exp\Big(\int_{\mathbb{T}} \frac{\zeta + z}{\zeta - z} \sum_{j=1}^{k} \log|u_{\alpha_j}(\zeta)| \, \mathrm{d}m(\zeta) \Big).$$

The series $\sum_j \log|u_{\alpha_j}|$ converges in $L^1(\mathbb{T})$, so g_k converges locally uniformly in $\mathbb{D}$ to an outer function. This function is f, since convergence in H^p implies locally uniform convergence in $\mathbb{D}$. □

6. A lower bound for the zeros

The proof of Theorem 3 rests on the following pointwise inequality, which controls the argument of the ratio of $1 - |E|^{p-2}E$ and $1 - E$.

Lemma 13. *Suppose that $1 < p < \infty$, and set $k = |p-2|/p$. For a complex number $E \neq 1$, let $K = |E|^{p-2}E$, where $K = 0$ if $E = 0$, and set $w = (1-K)/(1-E)$. Then $\operatorname{Re} w > 0$ and $|\operatorname{Im} w| \le k|w|$.*

Proof. If $E = 0$, then $w = 1$. Otherwise, write $E = te^{i\theta}$ with $t > 0$, and set $u = t^{p-1}$ and $c = \cos\theta$, so that $K = ue^{i\theta}$. Then

$$\operatorname{Re}\big((1-K)\overline{(1-E)}\big) = 1 + tu - (t+u)c \ge (1-t)(1-u) \ge 0,$$

since t and u lie on the same side of 1, and equality throughout forces $t = 1$ and $c = 1$, that is, $E = 1$. Hence $\operatorname{Re} w > 0$. Moreover, $\operatorname{Im}\big((1-K)\overline{(1-E)}\big) = (t-u)\sin\theta$, so the assertion is equivalent to

$$(t-u)^2(1-c^2) \le k^2|1-E|^2|1-K|^2.$$

Write $t = e^x$ and $q = p-1$, and set $s = \sinh((q-1)x/2)$ and $S = \sinh((q+1)x/2)$. Then $|1-E|^2 = 2t(\cosh x - c)$, $|1-K|^2 = 2u(\cosh qx - c)$ and $(t-u)^2 = 4tus^2$, so we need to show that

$$G(c) = k^2(\cosh x - c)(\cosh qx - c) - s^2(1-c^2) \ge 0.$$

The identities $\cosh qx - \cosh x = 2Ss$ and $\cosh x \cosh qx = 1 + S^2 + s^2$ show that the quadratic polynomial G has leading coefficient $k^2 + s^2 \ge 0$ and discriminant $4s^2(1-k^2)(s^2 - k^2S^2)$. Since $k = |q-1|/(q+1)$ and $y \mapsto \sinh(y)/y$ is even and increasing for $y > 0$, we have $|s| \le k|S|$. Hence the discriminant is nonpositive, and $G \ge 0$. □

In terms of (1), Lemma 13 compares the duality map $E \mapsto |E|^{p-2}E$ with the identity: the quotient of $1 - |E|^{p-2}E$ and $1 - E$ always lies in the sector $|\arg w| \le \arcsin k$, which is the half-line $w > 0$ when $p = 2$. The constant

k is optimal: letting $E \to 1$ along a suitable direction, the argument of w tends to $\arcsin k$. Armed with Lemma 13, we can now prove Theorem 3.

Proof of Theorem 3. Write the OPA as $q = (z-a)P$. For every $r \in \mathcal{P}_1$, the polynomial rP has degree at most n, so $z-a$ is a multiple of the OPA to $1/(Pf)$ of degree one. Replacing f by a constant multiple of Pf, we may therefore assume that $n = 1$ and that the OPA is $z-a$. Set $E = 1-(z-a)f$ and $K = |E|^{p-2}E$ on $\mathbb{T}$. Since $z-a$ is the OPA, the orthogonality relation (1) with $q' = z$ gives, after complex conjugation, $\int_{\mathbb{T}} K\overline{zf}\,\mathrm{d}m = 0$. Since zf vanishes at the origin, we also have $\int_{\mathbb{T}} \overline{zf}\,\mathrm{d}m = 0$, and so

$$\int_{\mathbb{T}} (1-K)\overline{zf}\,\mathrm{d}m = 0.$$

If $E = 1$ on a set of positive measure, then $f = 0$ there, which is impossible for a nonzero function in H^p. Hence $w = (1-K)/(1-E)$ is defined almost everywhere on $\mathbb{T}$. Inserting $1-K = w(1-E) = w(z-a)f$ and using that $(z-a)\overline{z} = 1-a\overline{z}$ on $\mathbb{T}$, we obtain an identity between integrable functions, since $|K| = |E|^{p-1}$ with $E \in L^p$ and $zf \in L^p$, so that $(1-K)\overline{zf} \in L^1$ by Hölder's inequality:

$$\int_{\mathbb{T}} w|f|^2(1-a\overline{z})\,\mathrm{d}m = 0. \tag{11}$$

Let $\beta = \arcsin(|p-2|/p)$, and suppose, for contradiction, that $|a| \le \cos\beta$. As z ranges over $\mathbb{T}$, the point $1-a\overline{z}$ ranges over the circle of radius $|a|$ centred at 1, so its argument is at most $\arcsin|a| \le \pi/2-\beta$ in absolute value, with equality at no more than two points. By Lemma 13, the argument of w is at most β in absolute value. Hence $\operatorname{Re}\bigl(w(1-a\overline{z})\bigr) > 0$ almost everywhere on $\mathbb{T}$. Since $|f| > 0$ almost everywhere on $\mathbb{T}$, the real part of the left-hand side of (11) is positive, a contradiction. Thus $|a| > \cos\beta = 2\sqrt{p-1}/p$. □

7. Two boundary peaks and the proof of Theorem 4

In this section we let the target concentrate at two conjugate points of $\mathbb{T}$. The limit problem then involves the following two functions of $C \in \mathbb{C}$. Let $H(t) = (1-it)^{-1}$ for $t \in \mathbb{R}$, and set

$$W_p(C) = \frac{1}{\pi C}\int_{\mathbb{R}} \bigl(1-|1-CH(t)|^{p-2}(1-CH(t))\bigr)\overline{H(t)}\,\mathrm{d}t \qquad (C \neq 0),$$

$$L_p(C) = \frac{1}{2\pi}\int_{\mathbb{R}} \bigl(|1-CH(t)|^p - 1 + p\operatorname{Re}\bigl(CH(t)\bigr)\bigr)\,\mathrm{d}t.$$

Both integrands are $O(t^{-2})$, since $|CH(t)| = O(1/|t|)$ and the integrand of L_p is $O(|CH(t)|^2)$. The integrand of L_p is nonnegative by convexity, and L_p is strictly convex, since $C \mapsto |1-CH(t)|^p$ is strictly convex for every t. Differentiating under the integral sign, we find that

$$\partial_{\overline{C}} L_p(C) = \frac{p}{4}\,C\,W_p(C). \tag{12}$$

The substitution $t \mapsto -t$ shows that $L_p(\overline{C}) = L_p(C)$ and $W_p(\overline{C}) = \overline{W_p(C)}$. Note also that with $E = 1 - CH(t)$ we have $\overline{H}/C = H\overline{H}/(1-E)$, so that

$$W_p(C) = \frac{1}{\pi}\int_{\mathbb{R}} \frac{1-|E|^{p-2}E}{1-E}\,\frac{\mathrm{d}t}{1+t^2}, \tag{13}$$

and by (13) and Lemma 13, $\operatorname{Re} W_p(C) > 0$.

Lemma 14. *Suppose that $1 < p < \infty$, and that $C \neq 0$ satisfies $\operatorname{Im} W_p(C) \neq 0$. Then*

$$\rho_1(p) \leq \frac{\operatorname{Re} W_p(C)}{|W_p(C)|}.$$

Proof. Step 1: the targets. Replacing C by $\overline{C}$ if necessary, we may assume that $W = W_p(C) = u + iv$ with $v < 0$. Set $a = u/|W|$, $s = -v/|W|$, $\zeta = -a + is$ and $\eta = C/(is)$. Then $\zeta \in \mathbb{T}$, $\operatorname{Im}\zeta > 0$, and $b = |\eta|^2 \operatorname{Re}(isW) = |C|^2|W| > 0$. Put $D = b - 2\operatorname{Re}\eta$, and for $0 < r < 1$ set $\delta = 1 - r$ and

$$f_r(z) = \delta\Big(D + \frac{\eta}{1-r\overline{\zeta}z} + \frac{\overline{\eta}}{1-r\zeta z}\Big).$$

The function f_r is rational with real Taylor coefficients, it is analytic in a neighborhood of $\overline{\mathbb{D}}$, and $f_r(0) = \delta b > 0$. Let $R_p(h) = |1-h|^p - 1 + p\operatorname{Re} h \geq 0$. Since $\int_{\mathbb{T}} Q f_r \,\mathrm{d}m = Q(0) f_r(0)$ for $Q \in \mathcal{P}_1$, the OPA to $1/f_r$ of degree one minimizes

$$J_r(Q) = \frac{\|1 - Qf_r\|_{H^p}^p - 1}{\delta} = -pb\operatorname{Re} Q(0) + \frac{1}{\delta}\int_{\mathbb{T}} R_p(Qf_r)\,\mathrm{d}m.$$

Step 2: the limit objective. We claim that, uniformly for Q in compact subsets of $\mathcal{P}_1$,

$$J_r(Q) \to J(Q) = -pb\operatorname{Re} Q(0) + L_p(\eta Q(\zeta)) + L_p(\overline{\eta} Q(\overline{\zeta})) \qquad (r \to 1^-).$$

Near ζ, substitute $z = \zeta e^{i\delta t}$, so that $\mathrm{d}m = \delta\,\mathrm{d}t/(2\pi)$ and $1 - r\overline{\zeta}z = \delta(1 - it) + O(\delta^2(1+t^2))$. Then $f_r(\zeta e^{i\delta t}) \to \eta H(t)$ locally uniformly in t, while the terms involving $\overline{\eta}$ and D are $O(\delta)$, since $\zeta \neq \overline{\zeta}$. The same holds near $\overline{\zeta}$ with η replaced by $\overline{\eta}$. For Q in a compact set, the functions Qf_r are uniformly bounded, so $R_p(Qf_r) \leq K|Qf_r|^2$, and $|f_r(e^{ix})| \leq K\delta/(\delta + |x - x_0|)$ near each of the two points $e^{\pm ix_0} = \zeta, \overline{\zeta}$. On arcs of a fixed length ε around the two points, dominated convergence with the majorant $K/(1+t^2)$ gives the two terms involving L_p. On the rest of $\mathbb{T}$ we have $|Qf_r| \leq K\delta/\varepsilon$, so its contribution to J_r is $O(\delta/\varepsilon^2)$.

Step 3: the OPAs are bounded. The minimizers of J_r are bounded as $r \to 1^-$. Since $\zeta \neq \overline{\zeta}$, there is $c > 0$ such that $\max(|Q(\zeta)|, |Q(\overline{\zeta})|) \geq cT$ for every $Q \in \mathcal{P}_1$ with coefficient norm T. If, say, $|Q(\zeta)| \geq cT$, then $|Q| \geq cT/2$ on a fixed arc around ζ, and $|f_r| \geq c' > 0$ on the arc of length δ around ζ for small δ. Using $R_p \geq 0$ elsewhere and $R_p(h) \geq c|h|^p - K|h| - K$ on that arc, we get $J_r(Q) \geq cT^p - KT - K$, with constants independent of r. Since $J_r(0) = 0$, the minimizers are bounded.

Step 4: the limit minimizer. By Lemma 1, the OPAs are characterized by an orthogonality relation; in the limit this relation becomes the vanishing of the derivatives of J, which we now compute. The function J is strictly convex and coercive, since $Q \mapsto (Q(\zeta), Q(\overline{\zeta}))$ is injective on $\mathcal{P}_1$. Since $L_p(\overline{C}) = L_p(C)$, we have $J(\overline{Q}) = J(Q)$, where $\overline{Q}$ has the conjugate coefficients, so the unique minimizer Q_* of J has real coefficients. We claim that $Q_*(z) = z + a$. Note that $\zeta + a = is$, so $\eta Q_*(\zeta) = C$. By (12), the derivative of $L_p(\eta Q(\zeta))$ at Q_* in a real direction $Q' \in \mathcal{P}_1$ equals $(p/2)\operatorname{Re}\bigl(\overline{CW}\eta Q'(\zeta)\bigr)$, and the second term of J contributes the same amount. For $Q' = 1$ we have $\overline{C}\eta = |C|^2/(is)$ and $\operatorname{Re}(\overline{W}/(is)) = -v/s = |W|$, so the derivative of J is $-pb + p|C|^2|W| = 0$. For $Q' = z$ we have $\zeta/(is) = 1 + ia/s$ and $\operatorname{Re}\bigl(\overline{W}(1 + ia/s)\bigr) = u + va/s = 0$, so the derivative vanishes as well. Hence $Q_* = z + a$.

Step 5: conclusion. By the uniform convergence and the boundedness of the minimizers, every limit point of the OPAs to $1/f_r$ of degree one minimizes J, so the OPAs converge coefficientwise to $z + a$. Their zeros converge to $-a$, and since $f_r(0) \neq 0$, we obtain $\rho_1(p) \leq |a| = \operatorname{Re} W/|W|$. □

Proof of Theorem 4. The upper bound follows from Theorem 3 and the expansion of $2\sqrt{p-1}/p$ recorded after it. For the lower bound, fix $C = 2 - 2i$ and write $E(t) = 1 - CH(t)$ and $\ell(t) = \log|E(t)|$. Since $|E(t)|^2 = (1 + (2 - t)^2)/(1 + t^2)$, the function $|E|$ is bounded above and away from 0 on $\mathbb{R}$, and $\ell(t) = O(1/|t|)$. Hence we may differentiate under the integral sign in p, with integrable majorants of order t^{-2}, locally uniformly in p. At $p = 2$ we have $W_2(C) = \pi^{-1}\int H\overline{H}\,\mathrm{d}t = 1$. Since $E\overline{H} = (1 + it)^{-1} - C(1 + t^2)^{-1}$, we get

$$\partial_p W_p(C)\big|_{p=2} = -\frac{1}{\pi C}\int_{\mathbb{R}} \frac{\ell(t)}{1 + it}\,\mathrm{d}t + \frac{1}{\pi}\int_{\mathbb{R}} \frac{\ell(t)}{1 + t^2}\,\mathrm{d}t.$$

The function $F(z) = \log\bigl((1 - i(z - 2))/(1 - iz)\bigr)$ is analytic in the upper half-plane, is $O(1/|z|)$ there, and has boundary real part ℓ. Its value at i is $Z = \log(1 + i) = (\log 2)/2 + i\pi/4$. The Poisson formula gives $\pi^{-1}\int \ell(t)(1 + t^2)^{-1}\,\mathrm{d}t = \operatorname{Re} Z$. Moreover, $\int F(t)(t - i)^{-1}\,\mathrm{d}t = 2\pi i Z$, while $\int \overline{F(t)}(t - i)^{-1}\,\mathrm{d}t = 0$, since $\overline{F(\overline{z})}$ is analytic in the lower half-plane; hence $\pi^{-1}\int \ell(t)(1 + it)^{-1}\,\mathrm{d}t = Z$. We conclude that

$$W_p(C) = 1 + (p - 2)\Bigl(\operatorname{Re} Z - \frac{Z}{C}\Bigr) + O\bigl((p - 2)^2\bigr),$$

and since $\operatorname{Im}(Z/C) = \kappa$, the argument of $W_p(C)$ is $-\kappa(p - 2) + O((p - 2)^2)$. In particular $\operatorname{Im} W_p(C) \neq 0$ for p close to 2, and Lemma 14 gives

$$1 - \rho_1(p) \geq 1 - \cos\bigl(\arg W_p(C)\bigr) = \frac{\kappa^2}{2}(p - 2)^2 + O\bigl(|p - 2|^3\bigr).$$

This proves the lower bound. □

8. The exact radius for $p = 4$

For $p = 4$ the duality map $E \mapsto |E|^2 E$ is a polynomial in E and $\overline{E}$, and the orthogonality relation of Lemma 1 can be analysed exactly. We first

move the zero to the origin by a Möbius transformation. For $w \in \mathbb{D}$ and $\xi \in \mathbb{T}$, let $S_w(\xi) = (1 - w\overline{\xi})^{-1}$, and write $\sigma = \operatorname{Re} S_w$ and $\tau = \operatorname{Im} S_w$. Then $\sigma > 0$, $P_w = 2\sigma - 1$ is the Poisson kernel at w, and $|\arg S_w| \leq \arcsin|w|$, since $1 - w\overline{\xi}$ lies on the circle of radius $|w|$ centred at 1. For $h \in \mathbb{C}$ we set $K(h) = |1 - h|^2(1 - h)$ and $g(h) = (1 - K(h))\overline{h}$.

Lemma 15. *Suppose that $f \in H^4$, that $a \in (0, 1)$, and that $z - a$ is a multiple of the OPA to $1/f$ of degree one in H^4. Then there is $H \in H^4$ with $H(0) = 0$ and $H \not\equiv 0$ such that $\int_{\mathbb{T}} g(H) S_{-a} \, \mathrm{d}m = 0$.*

Proof. Replacing f by a constant multiple, we may assume that $z - a$ is the OPA, and put $h = (z - a)f$. As in the proof of Theorem 3, the orthogonality relation (1) with $q' = z$ and the identity $\int_{\mathbb{T}} \overline{zf} \, \mathrm{d}m = 0$ give $\int_{\mathbb{T}} (1 - K(h))\overline{zf} \, \mathrm{d}m = 0$. On $\mathbb{T}$ we have $\overline{zf} = \overline{h}/(1 - az)$, so $\int_{\mathbb{T}} g(h)(1 - az)^{-1} \, \mathrm{d}m(z) = 0$. Substitute $z = (\xi + a)/(1 + a\xi)$. Then $1 - az = (1 - a^2)/(1 + a\xi)$ and $\mathrm{d}m(z) = (1 - a^2)|1 + a\xi|^{-2} \, \mathrm{d}m(\xi)$, so that $\mathrm{d}m(z)/(1 - az) = S_{-a}(\xi) \, \mathrm{d}m(\xi)$. Hence $H(\xi) = h\big((\xi + a)/(1 + a\xi)\big)$ has the required properties; $H(0) = h(a) = 0$, and $H \not\equiv 0$ since $f \not\equiv 0$. □

Since rotations do not affect the problem, Theorem 5 therefore follows from the next result, applied with $w = -a$.

Proposition 1. *Suppose that $H \in H^4$, that $H(0) = 0$, that $H \not\equiv 0$, and that w is real with $|w| \leq \sqrt{7/8}$. Then*

$$\operatorname{Re} \int_{\mathbb{T}} g(H) S_w \, \mathrm{d}m > 0.$$

Proof. Step 1: a sum of squares. Let $h \in \mathbb{C}$, $u = |h|^2$ and $T = S_w + 2\overline{S_w}$, so that $\operatorname{Re} T = 3\sigma$ and $\operatorname{Im} T = -\tau$. Expanding, we find that $g(h) = 2u + u^2 - uh - 2u\overline{h} + \overline{h}^2$ and $\operatorname{Re}\big((h + 2\overline{h})S_w\big) = \operatorname{Re}(Th)$. Completing the square in u gives, pointwise on $\mathbb{T}$,

$$\operatorname{Re}\big(g(h)S_w\big) = \sigma\Big(u - \frac{\operatorname{Re}(Th)}{2\sigma}\Big)^2 + \frac{D}{4} + \operatorname{Re}\big(S_w \overline{h}^2\big), \qquad D = 8\sigma u - \frac{\operatorname{Re}(Th)^2}{\sigma}.$$

The first term is nonnegative. Since $\overline{S_w}$ is analytic in ξ and $H(0) = 0$, we have $\int_{\mathbb{T}} S_w \overline{H}^2 \, \mathrm{d}m = \overline{H(0)^2} = 0$. Hence $\operatorname{Re} \int_{\mathbb{T}} g(H) S_w \, \mathrm{d}m \geq (1/4) \int_{\mathbb{T}} D(H) \, \mathrm{d}m$.

Step 2: a quadratic form. Write $H = x + iy$ on $\mathbb{T}$. Then $\operatorname{Re}(TH) = 3\sigma x + \tau y$, and a direct computation gives

$$D = \omega y^2 - \operatorname{Re}\big((\sigma - 3i\tau)H^2\big), \qquad \omega = \frac{7\sigma^2 - \tau^2}{\sigma}.$$

Since $\sigma - 3i\tau = -S_w + 2\overline{S_w}$, the reproducing property of S_w and $H(0) = 0$ give $\int_{\mathbb{T}} (\sigma - 3i\tau)H^2 \, \mathrm{d}m = -H(w)^2$. Consequently,

$$\int_{\mathbb{T}} D \, \mathrm{d}m = \int_{\mathbb{T}} \omega y^2 \, \mathrm{d}m + \operatorname{Re}\big(H(w)^2\big) \geq \int_{\mathbb{T}} \omega y^2 \, \mathrm{d}m - \big(\operatorname{Im} H(w)\big)^2.$$

Step 3: the Cauchy–Schwarz inequality. Since $|\tau|/\sigma \leq \tan \arcsin|w| \leq \sqrt{7}$ when $|w|^2 \leq 7/8$, we have $\omega \geq 0$, with equality at no more than two points.

Since $\int_{\mathbb{T}} y\,\mathrm{d}m = \operatorname{Im} H(0) = 0$, we have $\operatorname{Im} H(w) = \int_{\mathbb{T}} y(P_w - c)\,\mathrm{d}m$ for every real c, and the Cauchy–Schwarz inequality gives

$$\big(\operatorname{Im} H(w)\big)^2 \le M(w,c)\int_{\mathbb{T}} \omega y^2\,\mathrm{d}m, \qquad M(w,c) = \int_{\mathbb{T}} \frac{(P_w-c)^2}{\omega}\,\mathrm{d}m.$$

Moreover, $y \not\equiv 0$, since otherwise H would be a real constant, hence 0. So $\int_{\mathbb{T}} D\,\mathrm{d}m > 0$ as soon as $M(w,c) < 1$ for some c.

Step 4: the integrals. We take $c = 1$, and first suppose that $|w| < \sqrt{7/8}$. Put $q = \sqrt{7-8w^2}$, so that $1 - w^2 = (1+q^2)/8$, and let $\alpha = \sqrt{7} + i$. Then $7\sigma^2 - \tau^2 = \operatorname{Re}(\alpha S_w)\operatorname{Re}(\overline{\alpha} S_w)$, and so

$$\frac{1}{\omega} = \frac{1}{2\sqrt{7}}\Big(\frac{1}{\operatorname{Re}(\alpha S_w)} + \frac{1}{\operatorname{Re}(\overline{\alpha} S_w)}\Big).$$

Write $\xi = e^{i\theta}$, and let $\vartheta_0 \in (0, \pi/2)$ be the angle with $\cos\vartheta_0 = \sqrt{7/8}$. Since $S_w = (1 - w\xi)/|1 - w\xi|^2$, we have

$$\operatorname{Re}(\alpha S_w) = \frac{A_+(\theta)}{|1-w\xi|^2} \quad \text{and} \quad \operatorname{Re}(\overline{\alpha} S_w) = \frac{A_-(\theta)}{|1-w\xi|^2},$$

where $A_\pm(\theta) = \sqrt{7} - \sqrt{8}\,w\cos(\theta \pm \vartheta_0)$. We use the elementary integrals

$$\int_{\mathbb{T}} \frac{\mathrm{d}m}{A - B\cos\psi} = \frac{1}{\sqrt{A^2 - B^2}}, \qquad \int_{\mathbb{T}} \frac{\cos\psi\,\mathrm{d}m}{A - B\cos\psi} = \frac{1}{B}\Big(\frac{A}{\sqrt{A^2-B^2}} - 1\Big),$$

valid for $A > |B|$, with $A = \sqrt{7}$, $B = \sqrt{8}w$ and $\sqrt{A^2 - B^2} = q$. Since $P_w = (1-w^2)/|1-w\xi|^2$, we get

$$\int_{\mathbb{T}} \frac{P_w}{\omega}\,\mathrm{d}m = \frac{1-w^2}{2\sqrt{7}}\int_{\mathbb{T}}\Big(\frac{1}{A_+} + \frac{1}{A_-}\Big)\,\mathrm{d}m = \frac{1-w^2}{\sqrt{7}q} = \frac{1+q^2}{8\sqrt{7}q}.$$

Next, $|1 - w\xi|^2 = 1 + w^2 - 2w\cos\theta$, and with $\psi = \theta \pm \vartheta_0$ we have $\cos\theta = \cos\psi\cos\vartheta_0 \pm \sin\psi\sin\vartheta_0$, where the terms with $\sin\psi$ integrate to zero. Hence

$$\int_{\mathbb{T}} \frac{\mathrm{d}m}{\omega} = \frac{1}{\sqrt{7}}\Big(\frac{1+w^2}{q} - \frac{2w\cos\vartheta_0}{\sqrt{8}w}\Big(\frac{\sqrt{7}}{q} - 1\Big)\Big) = \frac{1}{4} + \frac{1-q^2}{8\sqrt{7}q}.$$

Finally, $\int_{\mathbb{T}} P_w^2/\omega\,\mathrm{d}m = \int_{\mathbb{T}} \mathrm{d}m/\omega$. To see this, let $\psi_w(\xi) = (w - \xi)/(1 - w\xi)$. Then $\int_{\mathbb{T}} F\,\mathrm{d}m = \int_{\mathbb{T}} (F \circ \psi_w) P_w\,\mathrm{d}m$ for integrable F, $P_w \circ \psi_w = 1/P_w$, and a direct computation gives $S_w \circ \psi_w = 1/((1-w^2)S_w)$, so that $\omega \circ \psi_w = \omega/P_w$. Applying the change of variables to $F = P_w^2/\omega$ gives the claim. Combining the three integrals,

$$M(w,1) = 2\int_{\mathbb{T}} \frac{\mathrm{d}m}{\omega} - 2\int_{\mathbb{T}} \frac{P_w}{\omega}\,\mathrm{d}m = \frac{1}{2} + \frac{1-q^2}{4\sqrt{7}q} - \frac{1+q^2}{4\sqrt{7}q} = \frac{\sqrt{7}-q}{2\sqrt{7}} \le \frac{1}{2}.$$

For $|w| = \sqrt{7/8}$, the integrand $(P_w - 1)^2/\omega$ is the pointwise limit of the corresponding integrands for $w' \to w$, and Fatou's lemma gives $M(w,1) \le 1/2$ as well. □

Proof of Theorem 5. By the reduction to degree one in the introduction, it suffices to consider $n = 1$, and by a rotation we may assume that the zero is $a \in (0, 1)$. If $a \le \sqrt{7/8}$, then Lemma 15 and Proposition 1 with $w = -a$ give a contradiction. Hence $|a| > \sqrt{7/8}$. The equality $\rho_1(4) = \sqrt{7/8}$ follows from Lemma 14: by (12), expanding $|1 - CH|^2(1 - CH)$ and integrating gives $W_4(C) = 2 - C/2 - \overline{C} + |C|^2/2$, and for $C = (3 - i\sqrt{7})/2$ we get $W_4(C) = (7 - i\sqrt{7})/4$ and $\operatorname{Re} W_4(C)/|W_4(C)| = \sqrt{7/8}$. □

The proof is sharp at every step for the targets of Lemma 14: in the limit the square in Step 1 vanishes, the Cauchy–Schwarz inequality in Step 3 becomes an equality, and ω vanishes at the two points where the targets concentrate. In particular, the constant $\sqrt{7/8}$ enters only through the condition $7\sigma^2 \ge \tau^2$.

9. Concluding remarks

9.1. **Two explicit profiles.** For $p = 4$ we have $\sigma = -1$ and $\gamma = 2$, and (6) can be evaluated exactly. Set $a = 2L_{1/2} = 1 - \lambda(1 - w^2)$. Then

$$\Psi'_{\phi_\lambda}(1/2) = \int_{\mathbb{T}} |1 - w|^2 |a|^2 (1/2 - \lambda \operatorname{Re}(w - w^2)) \, \mathrm{d}m.$$

Using $|1 - w|^2 = 2 - w - \overline{w}$ and $|a|^2 = (1 - \lambda)^2 + \lambda^2 + \lambda(1 - \lambda)(w^2 + \overline{w}^2)$ and reading off the constant Fourier coefficient, we find that

$$\Psi'_{\phi_\lambda}(1/2) = 1 - \lambda + 3\lambda^2 - \lambda^3.$$

The cubic $\lambda^3 - 3\lambda^2 + \lambda - 1$ has a single real zero $\lambda_0 \approx 2.769$, since its local maximum, attained at $\lambda = 1 - \sqrt{2/3}$, is negative. Hence $v_* > 1/2$ if and only if $\lambda > \lambda_0$. For instance, $\phi_3(w) = 4 - 6w + 3w^2$ has the zeros $1 \pm i/\sqrt{3}$ of modulus $\sqrt{4/3}$, and $\Psi'_{\phi_3}(1/2) = -2$.

For $1 < p < 4/3$, the profile $\phi(w) = 2 - w$, which does not belong to our family, also works. Here $R_\phi = -1$, $L_{1/2} = -w/2$ and $\operatorname{Re} h_\phi = 1/2 - \operatorname{Re} w = (|1 - w|^2 - 1)/2$, so (6) gives

$$\Psi'_\phi(1/2) = p 2^{1-p} (M(p) - M(p-2)), \qquad M(a) = \int_{\mathbb{T}} |1 - w|^a \, \mathrm{d}m = 2^a I(a, 0).$$

Since $M(p)/M(p-2) = 4(p-1)/p$, this is negative precisely when $1 < p < 4/3$. The corresponding targets are $f_\alpha(z) = (2 - \alpha + (2\alpha - 1)z)/(1 + \alpha z)^2$.

9.2. **A certified example.** The proof of Theorem 1 does not produce a specific target, but for $p = 4$ an example can be verified in exact rational arithmetic. Let

$$\begin{aligned} g(z) = 10 - 15z + 22z^2 - 29z^3 + 36z^4 - 41z^5 + 44z^6 - 45z^7 + 44z^8 \\ - 41z^9 + 36z^{10} - 30z^{11} + 23z^{12} - 17z^{13} + 12z^{14} - 7z^{15} + 4z^{16}. \end{aligned}$$

The Schur–Cohn test, carried out with exact arithmetic, shows that g does not vanish in $\overline{\mathbb{D}}$; numerically, its zeros have modulus at least 1.030. Since g

has real coefficients, the uniqueness of the OPA shows that the OPA to $1/g$ of degree one is $A + Bz$ with A, B real, and that (A, B) minimizes

$$\Phi(A, B) = \|1 - (A + Bz)g\|_{H^4}^4$$

over $\mathbb{R}^2$. Note that $\Phi(A, B)$ is the sum of the squares of the coefficients of $\big(1-(A+Bz)g\big)^2$, so Φ is a convex polynomial with rational coefficients. The OPA vanishes in $\mathbb{D}$ if and only if $|A| < |B|$.

The set where $|A| \geq |B|$ is the union of two closed convex cones, bounded by the lines $A = \pm B$. Suppose that P_0 is a point at which Φ is smaller than its minimum on both lines. If (A, B) were in one of the cones, then the segment from (A, B) to P_0 would meet one of the lines at a point Y, and convexity would give $\Phi(Y) \leq \Phi(P_0)$, which is impossible. For $P_0 = (79/5000, 1583/100000)$ we find that $\Phi(P_0) < 0.734292$. On the lines we have

$$\Phi(t, t) = 1 - 40t + 2352t^2 - 86624t^3 + 1939246t^4,$$
$$\Phi(t, -t) = 1 - 40t + 244896t^2 - 93938680t^3 + 36668072718t^4.$$

If h is convex and $h'(t_1) < 0 < h'(t_2)$, then $h \geq h(t_1) + h'(t_1)(t_2 - t_1)$ on $\mathbb{R}$. Using this with $(t_1, t_2) = (157/10000, 79/5000)$ and $(t_1, t_2) = (0, 1/1000)$, respectively, we find that $\Phi(t, t) > 0.734323$ and $\Phi(t, -t) \geq 0.96$ for every real t. Hence $|A| < |B|$, and the OPA to $1/g$ of degree one vanishes in $\mathbb{D}$. These computations, including the Schur–Cohn test, are reproduced in exact arithmetic by the script `numerics/certificate.py` accompanying this paper. Numerically, its zero is $-A/B \approx -0.99807$.

The polynomial g was found numerically, by truncating the Taylor series of a target of the form f_α and optimizing its coefficients. A certificate of the same kind exists for a polynomial of degree 14, but we have not found an example of lower degree.

9.3. **The size of the parameters.** Our proof does not quantify how large λ must be, nor how close α must be to 1. A numerical evaluation of (6) suggests that $\Psi'_{\phi_\lambda}(1/2) < 0$ precisely for λ above a threshold $\lambda_0(p)$ with $\lambda_0 \approx 7.3$, 37 and 197 for $p = 1.8$, 1.95 and 1.99, and $\lambda_0 \approx 202$, 42, 12, 3.9 and 2.77 for $p = 2.01$, 2.05, 2.2, 3 and 4. Near $p = 2$ this is close to $2/|p-2|$, in accordance with Corollary 2: the derivative equals 1 at $p = 2$, while the leading term in Lemma 9 is proportional to $|\gamma|\lambda^{p-1}$. The zeros of the resulting OPAs lie very close to $\mathbb{T}$, at distance of order $1 - \alpha$.

The simpler profile $1 + \lambda(1 - w)$ is far less efficient. Its analogue of F_σ is $\int_{\mathbb{T}} |1 - w^2|^\gamma \operatorname{Re} w \, \mathrm{d}m$, which vanishes by the substitution $w \mapsto -w$, so the sign of the derivative is decided by lower-order terms.

9.4. **Inner factors and unimodular constants.** Let $q = a(z - z_0)$ with $z_0 \in \mathbb{D}$ be an OPA of degree one as in Theorem 1, and put

$$B(z) = \frac{z - z_0}{1 - \overline{z_0} z} \qquad \text{and} \qquad F = a(1 - \overline{z_0} z)g.$$

Then F is outer with $F(0) \neq 0$, and $\|1 - BF\|_{H^p} = \|1 - qg\|_{H^p} \leq \|1 - cF\|_{H^p}$ for every $c \in \mathbb{C}$, since $ca(1 - \overline{z_0}z)$ is in $\mathcal{P}_1$. Hence the strict inequality $\|1 - BF\|_{H^p} > \|1 - c_F F\|_{H^p}$ predicted in [1, Conjecture 4.3] fails for the pair (F, B), for every $p \neq 2$.

9.5. **Many degrees.** Some irregularity of the target in Theorem 2 is necessary. Suppose that f is analytic and does not vanish in a neighborhood of $\overline{\mathbb{D}}$, and let q_n and T_n denote the OPA to $1/f$ of degree n and the Taylor polynomial of $1/f$ of degree n. Then $\|1 - q_n f\|_{H^p} \leq \|1 - T_n f\|_{H^p} \leq K\rho^n$ for some $\rho < 1$, so $\|q_n - T_n\|_{H^p} \leq K\rho^n$, and Lemma 2 (b) gives $\|q_n - T_n\|_\infty \leq Kn^{1/p}\rho^n$. Hence $q_n \to 1/f$ uniformly on $\overline{\mathbb{D}}$, and q_n does not vanish in $\overline{\mathbb{D}}$ for large n. In our construction, the factors u_{α_k} have poles at $-1/\alpha_k$, which accumulate at -1, and the zeros of the OPAs of degree $N \in [a_k, b_k]$ are created at the scale $1 - \alpha_k$ around -1. We do not know whether f is bounded. This is consistent with the fact that, since f is outer, $q_n f \to 1$ in H^p, so the zeros of q_n eventually leave every compact subset of $\mathbb{D}$.

On the other hand, applying Lemma 12 with $g = 1$ and then Lemma 10 to the Taylor polynomials of u_α, we find that for every M there is a polynomial g which does not vanish in $\overline{\mathbb{D}}$ and whose OPAs of M consecutive degrees all have a zero in $\mathbb{D}$.

Problem 1. *Suppose that $p \neq 2$. Is there a bounded outer function f such that the OPA to $1/f$ of degree n has a zero in $\mathbb{D}$ for infinitely many n? Is there an outer function f for which this holds for every sufficiently large n?*

9.6. **How deep can the zeros be?** For $p \neq 2$ and $n \geq 1$, let $\rho_n(p)$ be defined as $\rho_1(p)$ in the introduction, with OPAs of degree n. By the reduction to degree one, $\rho_n(p) \geq \rho_1(p)$. For $p = 4$, the targets of Lemma 14 with $C = (3 - i\sqrt{7})/2$ are

$$f_r(z) = (1 - r)\,\frac{2\sqrt{14} + 18rz + 4\sqrt{14}\,r^2z^2}{1 + (\sqrt{14}/2)rz + r^2z^2},$$

and their OPAs of degree one have zeros tending to $-\sqrt{7/8}$ as $r \to 1^-$ (numerically, of modulus 0.9953, 0.9628 and 0.9401 for $r = 0.99$, 0.995 and 0.999). These targets are not outer for $r > 1/\sqrt{2}$.

For other p, the proof of Proposition 1 does not apply, since the duality map is not a polynomial. Optimizing C in Lemma 14 numerically gives the upper bounds $\rho_1(3) \leq 0.9753$, $\rho_1(5/2) \leq 0.9920$ and $\rho_1(6) \leq 0.8613$. Near $p = 2$ the constant in Theorem 4 can be improved by an explicit choice of C. Let $\tau_* \approx 0.94775$ solve $\sin 2\tau_* = \tau_*$, and put $R = 2\exp(1 - \tau_* \cot \tau_*)$, $C_* = Re^{-i\tau_*} = x + iy$ and $k_* = \sin \tau_*/R$. Since $x > 1$, the function $F(z) = \log\big((x - 1 - i(z + y))/(1 - iz)\big)$ is analytic in the upper half-plane, and the proof of Theorem 4 applies with $Z = F(i) = \log(\overline{C_*}/2)$. A direct computation gives $\operatorname{Im}(Z/C_*) = k_*$, and hence $\liminf_{p\to 2}(1 - \rho_1(p))/(p - 2)^2 \geq k_*^2/2 \approx 0.04355$. This choice of C maximizes $\operatorname{Im}(Z/C)$ over $\operatorname{Re} C > 1$; whether $k_*^2/2$ is the true limit is the content of the following conjecture.

Conjecture 1. *For every $p \neq 2$, $\rho_1(p) = \cos \sup_C |\arg W_p(C)|$. In particular, $(1-\rho_1(p))/(p-2)^2 \to k_*^2/2$ as $p \to 2$.*

Problem 2. *Are the zeros of OPAs to $1/f$ for outer f as deep as for general f? Does $\rho_n(p) = \rho_1(p)$ hold for every n?*

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